

A decorative graphic on the left side of the slide consisting of two overlapping parallelograms. The front one is blue and the back one is a light greenish-blue. They are positioned diagonally, with the blue one partially covering the green one.

Data Science Project Stage II

Group 8

Team Notebook - Weekly Stats across US

- Average of ~614,000 new cases per week
- Average of ~8,000 deaths per week
- By the end of our date range, there were:
 - Over 18,460,000 total cases
 - Almost 330,000 deaths
- The worst weeks for cases and deaths:
 - Almost 1,500,000 new cases in one week
 - Almost 19,000 deaths in one week

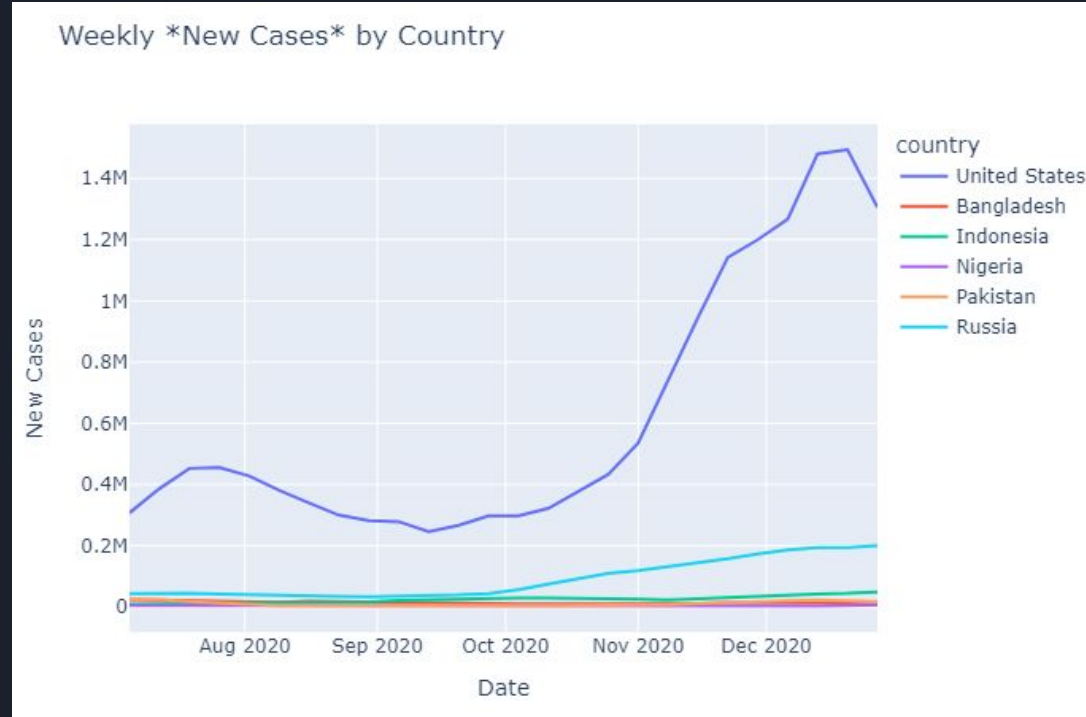
	total_cases	total_deaths	new_cases	new_deaths
count	27	27	26	26
mean	8075873	206648	614069	7965
std	4397228	55974	424329	4193
min	2480211	122768	246029	4571
25%	5062301	165437	301750	5349
50%	6903428	202226	407518	6432
75%	10081476	238090	894714	7833
max	18446012	329871	1494760	18893

Team Notebook - Compare Stats

Weekly New Cases Before Normalization

Note: Decentralized COVID Measures

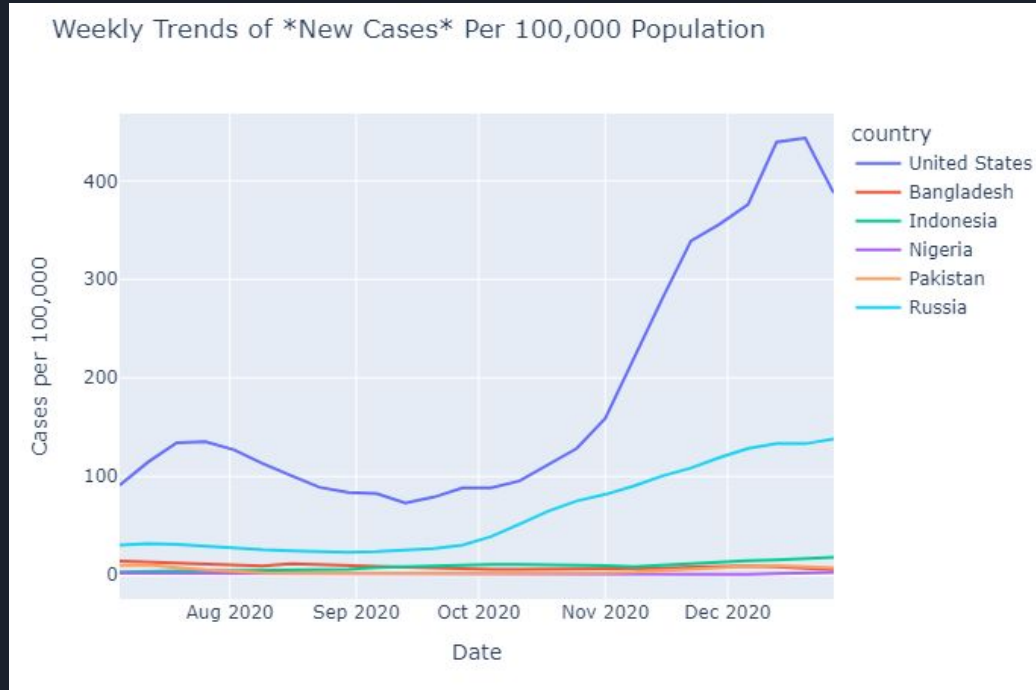
- Bimodal Distribution
- “Opening Up America Again” - April 16
- US Election Season ~Oct. / Nov.
- Holiday Season ~Nov. / Dec.
- Restriction Fatigue



Team Notebook - Compare Stats

Weekly New Cases After Normalization for Population

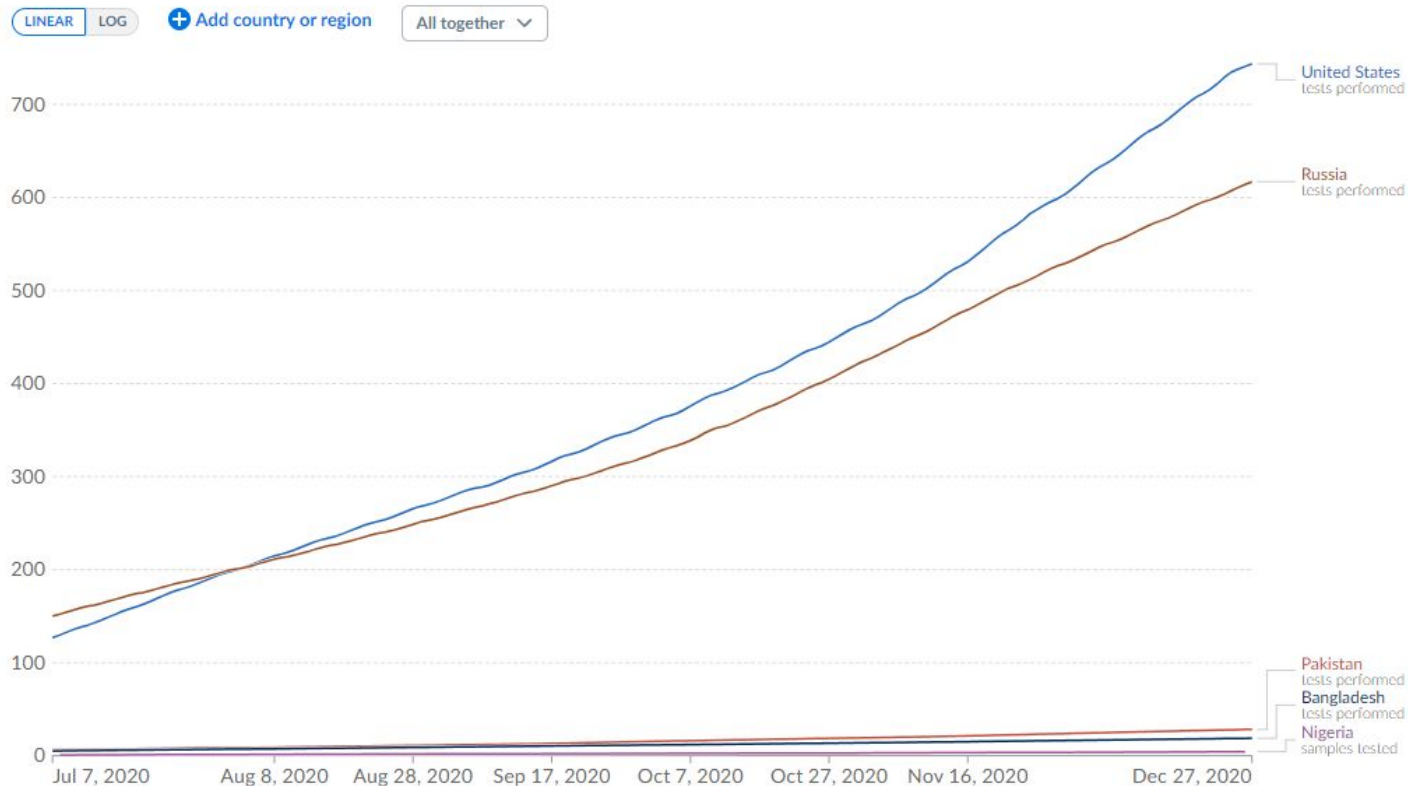
- Even per capita, US spread seems much higher
- Better access to testing



Total COVID-19 tests per 1,000 people

Comparisons across countries are affected by differences in testing policies and reporting methods.

Our World
in Data



Source: Official sources collated by Our World in Data

OurWorldInData.org/coronavirus • CC BY

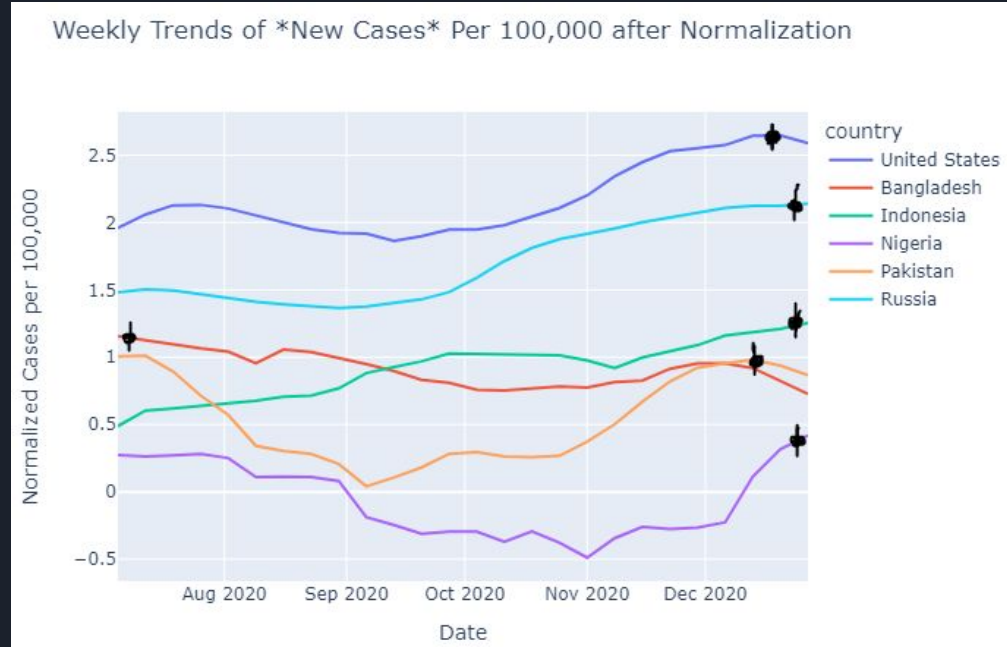
▶ Mar 1, 2020 Jun 23, 2022

Team Notebook - Compare Trends

Weekly New Cases Per 100,000, Log Normalized

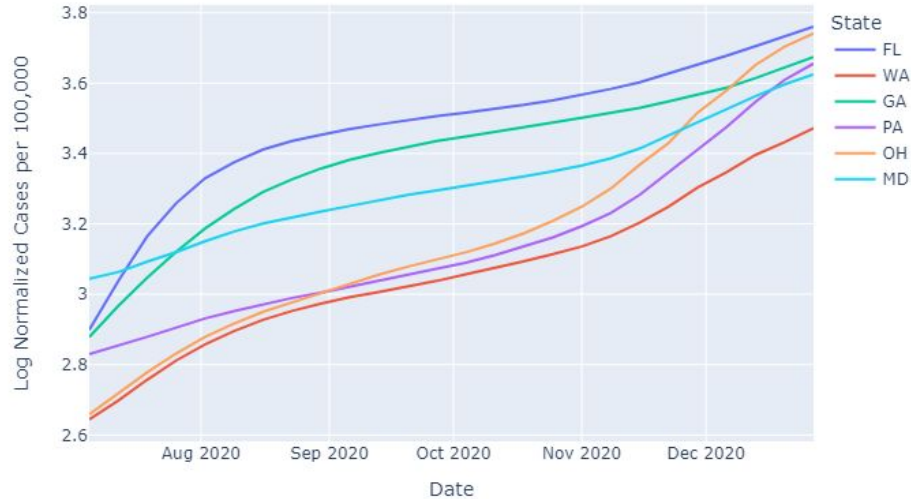
- US and Russia followed very similar trends
- Every Country saw rises around December
- Nigeria Data may be incomplete
- Bangladesh saw worse initial peak
- Pakistan saw significant improvement early

Death Trends generally correlated with case trends



Individual Notebook - Jesse Carter

Weekly Log Normalized *Cumulative Cases* Per 100,000 Population



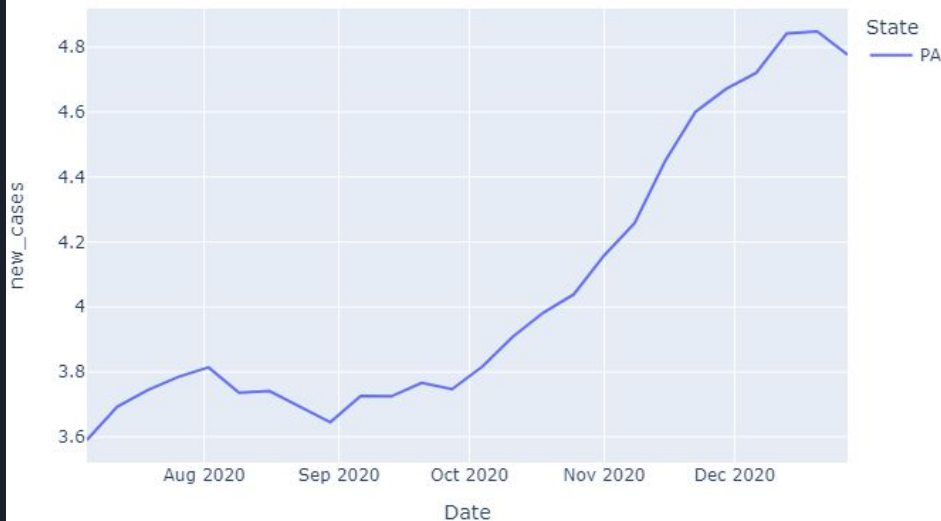
Weekly Log Normalized *New Cases* Per 100,000 Population



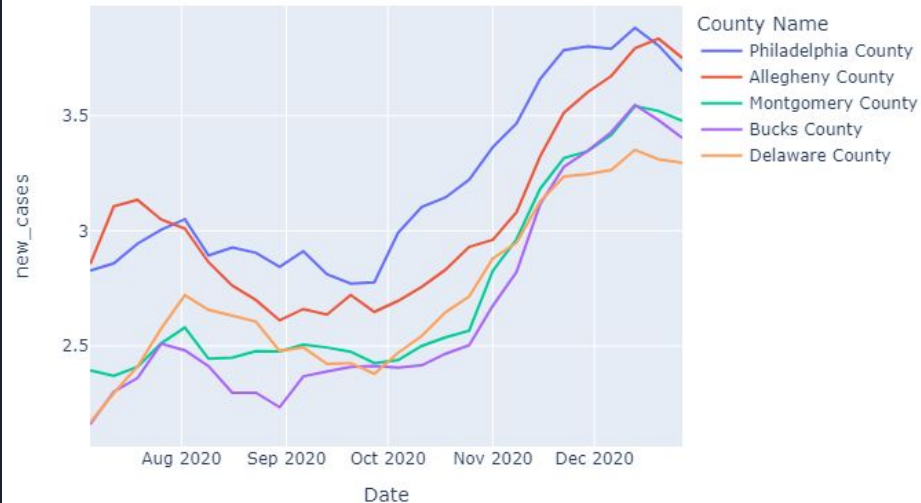
- WA consistently had the lowest total cases per capita
 - Aggressive Covid Restrictions
- FL had the highest total cases per capita
 - Tourism

Individual Notebook - Jesse Carter

Weekly *New Cases* Log Normalized



Weekly Log Normalized *New Cases* by County



- Despite lower infection rates than the US on average, PA saw higher death rates
- Older population on average

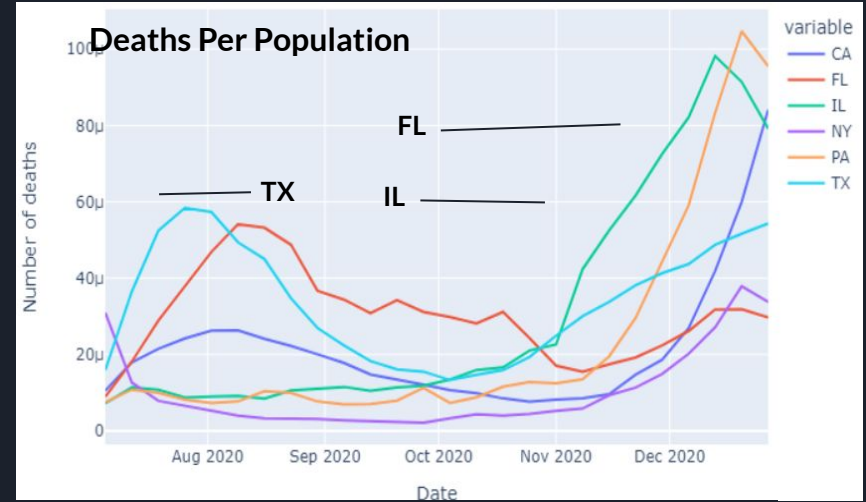
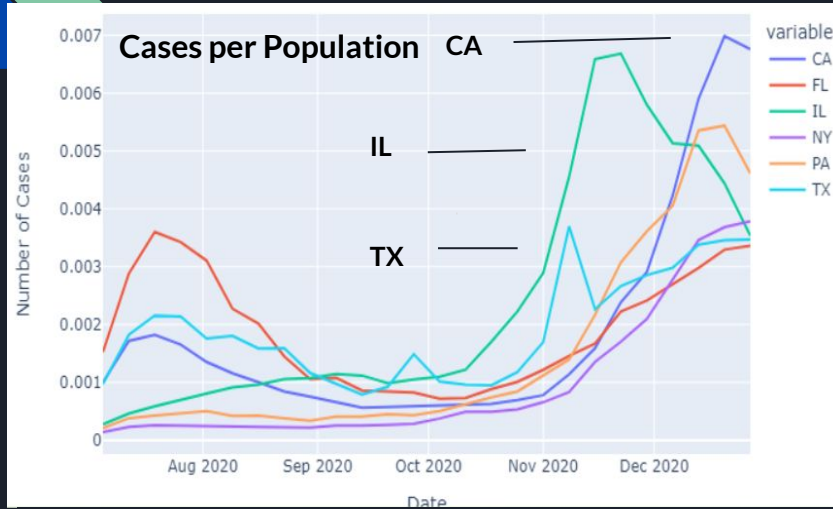
Individual Notebook-Sumithra Hariguruprasad

Weekly trends in number of cases and deaths between the period from July to December



- ❖ Texas has 254 unique counties
- ❖ Harris, Dallas, Tarrant, Bexar & El Paso - top 5 counties in highest number of cases.
- ❖ Harris, Dallas, El Paso Hidalgo & Bexa - top 5 counties in highest number of deaths

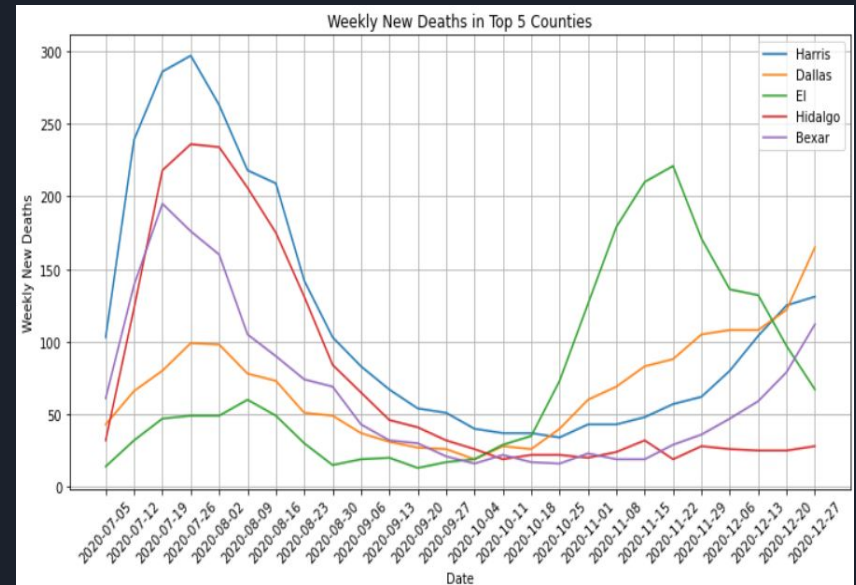
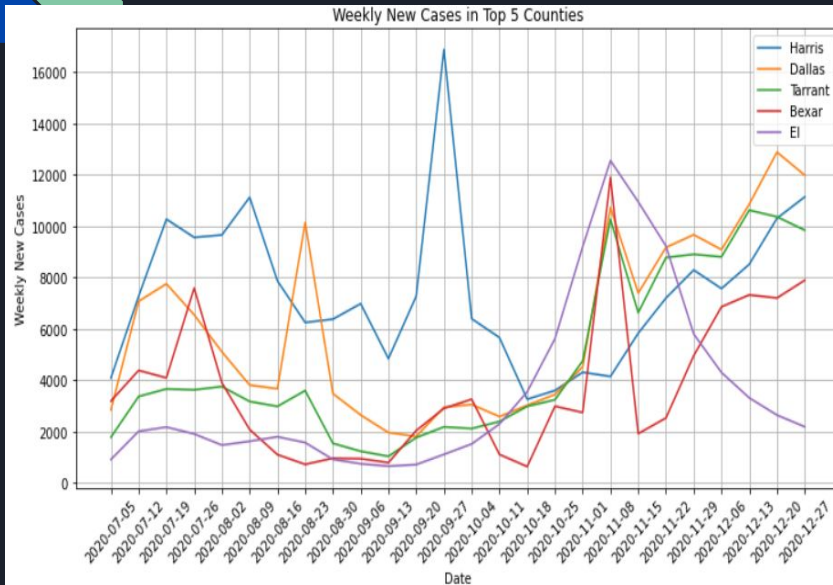
Weekly Trends in Texas and Other States



Texas being second most populous state in the US, it's compared with other top 6 most populous states (California, Texas Florida, new york and Pennsylvania) in terms of number of COVID cases and deaths. Despite its large population,

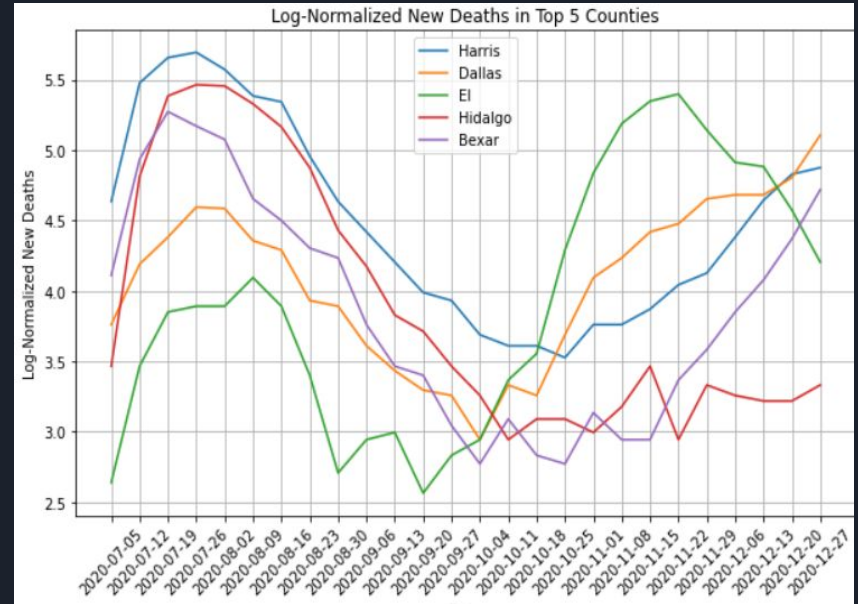
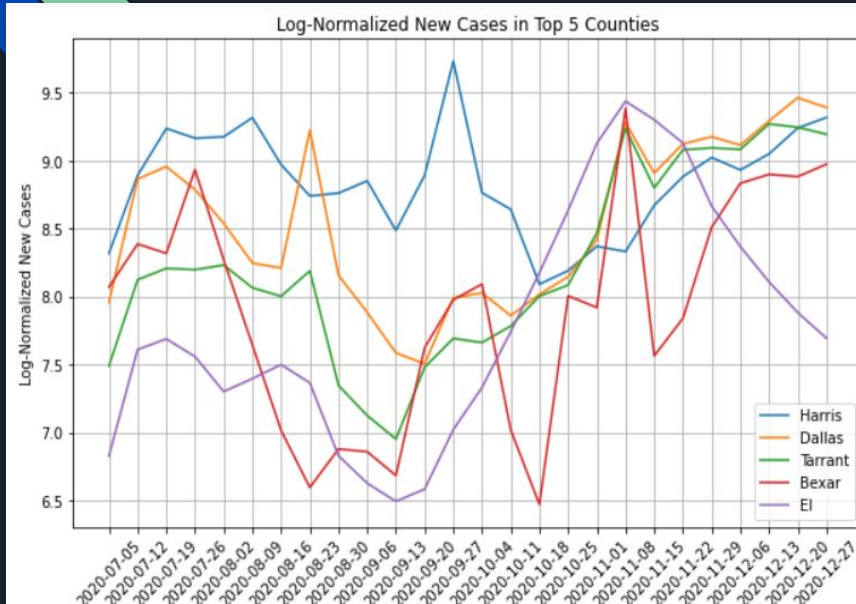
- ❖ The total number of cases in TX are relatively lesser than FL, CA, IL and NY during the last week of Nov and in 1st week Dec.
- ❖ The total number of deaths consistently stayed lesser than IL and FL from the last week of Nov and lesser than CA in the last 2 weeks of Nov.

Weekly Trends in Texas Counties



- ❖ All The counties in Texas experienced inconsistencies in COVID cases.
- ❖ Harris County had its peak in the 3rd week of Sep.
- ❖ El Paso and Bexar saw their highest case counts in 1st the week of November.
- ❖ Dallas saw the worst peak in 3rd week of August and Tarrant in 1st week of December
- ❖ All counties experienced a similar pattern of peaking and dropping in case counts at the same time.

Weekly Trends in Counties After Log Normalized



- ❖ Texas' largest counties — Harris, Dallas, Tarrant, Bexar and Travis — have the highest proportions of Black and Hispanic populations who were badly hit by COVID-19.
- ❖ Reasons for the peaks in different counties corresponds to the reasons stated for the US trends!